AUTOMATING DATA IMPACT & RELATIONSHIP MAPPING

Project Documentation Template

Date	07-April-2026
Team ID	SWTID-2026-2633
Project Name	Automating data import and relationship mapping and dot walking
Maximum Marks	

1. INTRODUCTION

Project Overview

This project focuses on building a system that automatically identifies, maps, and visualizes relationships between datasets, tables, and data fields across systems. It helps organizations understand how data flows and how changes in one dataset impact others.

Purpose

- Reduce manual effort in data lineage tracking
- Improve data governance and compliance
- Enable faster impact analysis during schema or data changes
- Support decision-making with clear data relationships

Abstract

- This project focuses on automating the process of data import and relationship mapping using Import Sets in ServiceNow. The system enables efficient loading of external data (such as CSV/Excel files) into staging tables and transforms it into target tables using Transform Maps.

Methodology

The project was completed using the following step-by-step methodology:

1. **Requirement Analysis**
    - Identified data sources and target tables
    - Defined relationship requirements (e.g., parent-child, CI relationships)

2. **Data Preparation**
    - Created sample datasets in CSV/Excel format
    - Cleaned and structured data for import
3. **Import Set Creation**
    - Loaded data into staging tables using Import Sets
    - Verified successful data ingestion
4. **Transform Map Configuration**
    - Mapped source fields to target fields
    - Applied transformation scripts where needed
5. **Relationship Mapping Logic**
    - Configured reference fields
    - Implemented scripts to establish relationships between records
6. **Testing & Validation**
    - Performed unit testing and UAT
    - Validated data accuracy and relationships
7. **Error Handling**
    - Identified duplicate or failed records
    - Implemented error logging and correction mechanisms

Implementation

The following components were developed:

- **Import Set Tables**
  Used to temporarily store incoming data from external sources
- **Transform Maps**
    - Automated transformation of data from staging to target tables
    - Included field mappings and scripts
- **Automated Data Import Process**
    - Enabled bulk data upload using CSV files
    - Reduced manual data entry

- **Relationship Mapping System**
  - Established links between records (e.g., parent-child relationships)
  - Ensured referential integrity
- **Error Handling Mechanism**
  - Identified invalid or duplicate records
  - Generated logs for failed transformations
- **Test Cases & Validation Reports**
  - Verified correctness of data import and relationships

References (Sources Used)

- Official documentation of ServiceNow
- ServiceNow Developer Portal
- Online tutorials and community forums
- Technical blogs on Import Sets and Transform Maps
- Course materials / class notes

  https://github.com/prakashchitra616-cyber/Automating-data-import-and-relationship-mapping-import-sets.git

2. IDEATION PHASE

Problem Statement

Organizations struggle to track how data moves across systems. Manual mapping is:

- Time-consuming
- Error-prone
- Difficult to maintain

Empathy Map Canvas

User Type	Thinks	Feels	Says	Does
Data Engineer	Where is this field used?”	Frustrated	Need lineage quickly”	Checks multiple systems
Analyst	Will this change affect reports?”	Uncertain	Is my dashboard safe?”	Is my dashboard safe?”
	“Is data reliable?”	Concerned	We need accuracy	Reviews reports

Brainstorming

Possible ideas:

- Metadata scanning tools
- Automated lineage graph generation
- AI-based relationship detection
- Impact alert system
- Visualization dashboards

3. REQUIREMENT ANALYSIS

Customer Journey Map

1. User logs into system
2. Uploads dataset / connects database
3. System scans metadata
4. Relationships are identified
5. Impact graph is generated
6. User analyzes dependencies

Solution Requirements

Functional Requirements:

- Data ingestion from multiple sources
- Automatic relationship detection
- Impact analysis engine
- Visualization dashboard

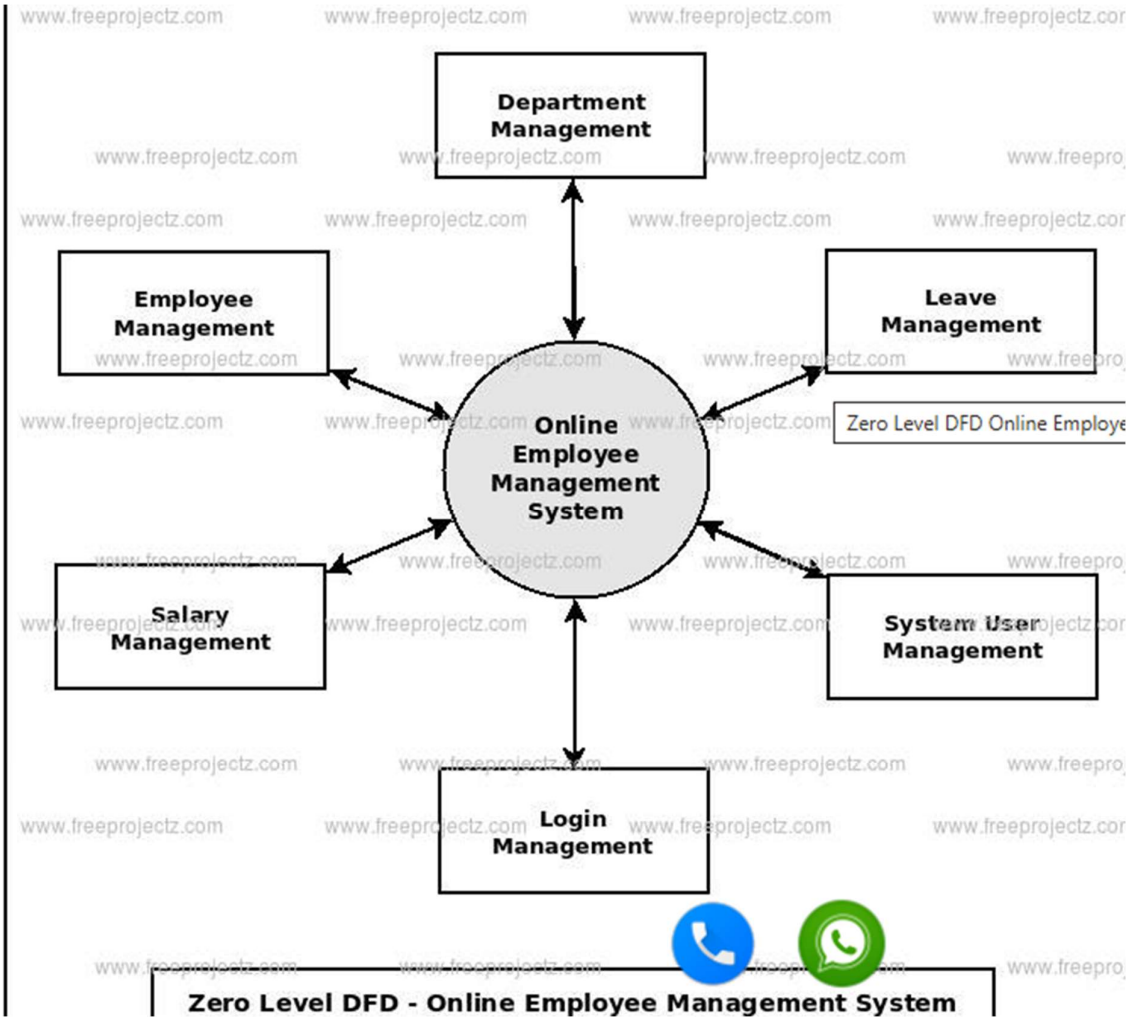
Non-Functional Requirements:

- High performance
- Scalability
- Security (data privacy)
- Reliability

Data Flow Diagram (DFD)

Flow:

Data Source → Data Ingestion → Processing Engine → Relationship Mapping → Visualization Dashboard



Technology Stack

Layer	Technology
Frontend	React / Angular
Backend	Python / Node.js
Database	PostgreSQL / MongoDB
Processing	Pandas / Apache Spark
Visualization	Power BI / D3.js
Cloud	AWS / Azure

4. PROJECT DESIGN

Problem-Solution Fit

Problem	Solution
Manual lineage tracking	Automated mapping
Data inconsistency	Centralized system
Slow impact analysis	Real-time processing

Proposed Solution

A system that:

- Automatically scans datasets
- Detects relationships using metadata and rules
- Generates impact paths
- Displays them in a visual graph

Solution Architecture

Components:

- Data Connector Layer
- Processing Engine
- Relationship Mapper
- Impact Analyzer
- Visualization Layer

5. PROJECT PLANNING & SCHEDULING

Project Planning

Phase	Task	Duration
Phase 1	Requirement Analysis	1 Week
Phase 2	Design	1 Week
Phase 3	Development	3 Weeks
Phase 4	Testing	1 Week
Phase 5	Deployment	1 Week

6. FUNCTIONAL AND PERFORMANCE TESTING

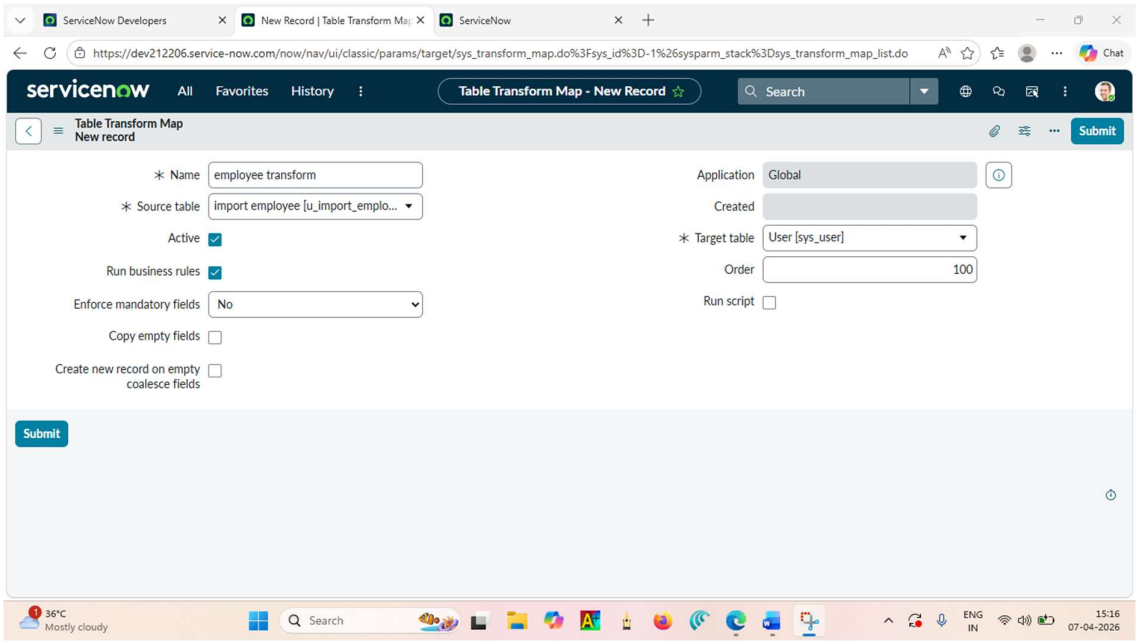
Performance Testing

- Load testing for large datasets
- Response time measurement
- Accuracy of relationship detection
- Scalability testing

7. RESULTS

Output Screenshots

/



ServiceNow Developers | employee transform | Table Transf | ServiceNow

https://dev212206.service-now.com/now/nav/ui/classic/params/target/sys_transform_map.do%3Fsys_id%3D4eec18ee8340031094dbe270ceaad383

servicenow All Favorites History Table Transform Map - employee tra... Search

Table Transform Map
employee transform

Copy Update Delete

* Name: employee transform
* Source table: import employee [u_import_empl...
Active: ☑
Run business rules: ☑
Enforce mandatory fields: No
Copy empty fields: ☐
Create new record on empty coalesce fields: ☐

Application: Global
Created: 2026-04-07 02:46:57
* Target table: User [sys_user]
Order: 100
Run script: ☐

Copy Update Delete

Related Links
Auto Map Matching Fields
Mapping Assist
Validate Coalesce Fields
Transform
Index Coalesce Fields
Run Point Scan

Field Maps (5) | Transform Scripts | Unindexed reference fields (1) | Empty reference fields (2)

36°C Mostly cloudy | Search | ENG IN | 15:19 07-04-2026

ServiceNow Developers | ServiceNow | ServiceNow

https://dev212206.service-now.com/now/nav/ui/classic/params/target/import_status.do%3Fsysparm_pworker_sysid%3Dba6e9ca28380031094dbe270ceaad3d3%...

servicenow All Favorites History Workspaces ServiceNow Search

Progress

Name: Transforming: ISET0010001
State: Complete
Completion code: Success
Message: Transformation complete

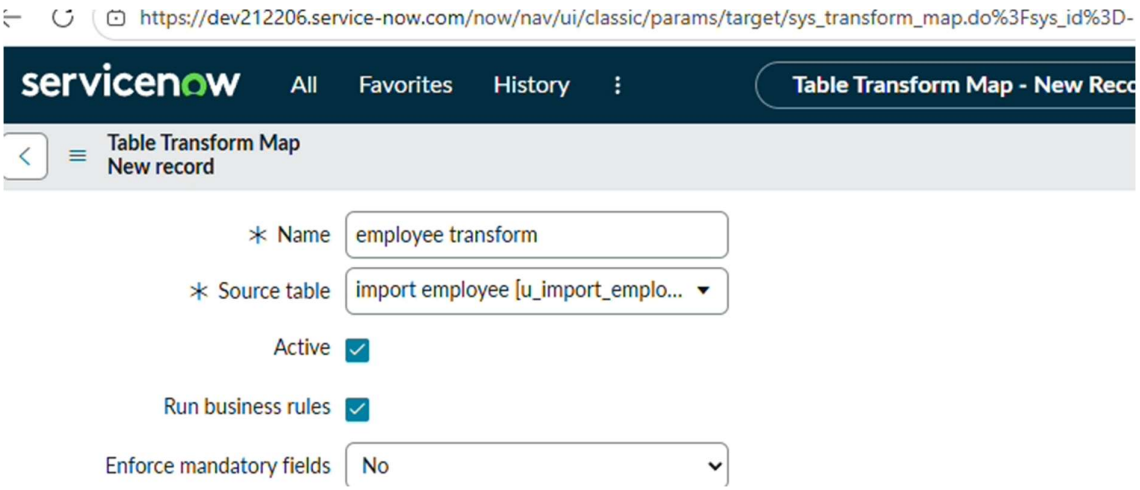
Next steps...

ISET0010001 Go to the import sets for this data load

Transform history Show the transform history, related errors and log

Import log View the import log

36°C Mostly cloudy | Search | ENG IN | 15:20 07-04-2026



8. ADVANTAGES & DISADVANTAGES

Advantages:

- Saves time
- Improves accuracy
- Better data governance
- Real-time insights

Disadvantages:

- Initial setup complexity
- Requires clean metadata
- High processing cost for large systems

9. CONCLUSION

The project successfully automates data relationship mapping and impact analysis, reducing manual effort and improving decision-making efficiency.

10. FUTURE SCOPE

- AI-based predictive impact analysis
- Integration with more data sources
- Real-time alerts for schema changes
- Advanced visualization with interactive graphs

11. APPENDI

Dataset Link

"C:\Users\ELCOT\Downloads\employee_data.csv"

GitHub & Project Demo Link

https://github.com/prakashchitra616-cyber/Automating-data-import-and-relationship-mapping-import-sets.git